

SGGS Institute of Engineering and Technology, Vishnupuri, Nanded. ISE-1 Examination, Academic Year 2025-26 (Sem1)
Class: Third Year (Electronics and Telecommunication Engineering)

Date: 23/08/25

Time: 15 to 15:45 p.m.

Subject: Microcontrollers and Applications, Subject code: PCC-EC 301
Max. Marks : 20 , Duration : 45 minutes

Note: I. Use of calculator is not allowed II. All questions are compulsory to solve.

Sr. No.	Questions	CO	Marks
Q.1	State difference between microprocessor and microcontroller? Why is the crystal oscillator frequency of 8051 microcontroller based system is 11.0592 MHZ? Write 8051 assembly language program to load the accumulator with the value FFH and complement the Accumulator 100 times.	CO1	05
Q.2	Discuss all registers of 8051 Microcontroller. Show the contents of PSW after execution of the following instructions, MOV A, #0BFH ADD A, #2BH	CO2	05
Q.3	Explain PUSH, DA, DJNZ and MOVX instructions.	CO1	05
Q.4	Write a 8051 assembly language program to add the values F1H, C1H, FFH, 10H and 10H. Explain direct and indirect addressing mode of 8051 Microcontroller.	CO2	05
	END		

CO1: Interpret architecture of 8-bit microprocessor and microcontrollers. CO2: Develop program in 8051 in assembly language for the given operation. CO3: Develop program using timers and interrupts. CO4: Interface the memory and I/O peripherals to 8051 microcontrollers. CO5: Design microcontroller-based applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electronics and Telecommunication Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY B.Tech	Course: Control Systems	Course Code: PCC-EC302/PCC-EC303
Date: 22.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	The control system is represented by following block diagram a) Determine the transfer function by using block diagram reduction technique. b) Draw the equivalent signal flow graph	09	CO1	BT2, BT3
Q.2	Find the transfer function for the block diagram given below using block diagram reduction technique	07	CO1	BT2, BT3
Q.3	a) Explain what is meant by open loop transfer function in relation to a closed loop control system. b) Determine the time domain function corresponding to pole location $s = 3 \pm j1$.	04	CO2	BT1, BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Model a physical system and express its internal dynamics and input-output relationships by means of block diagrams, mathematical model and transfer functions.
CO2	Understand and explain the relationships between the parameters of a control system and its stability, accuracy, transient behaviour.
CO3	Identify the parameters that the system is sensitive to. Determine the stability of a system and parameter ranges for a desired degree of stability.
CO4	Plot the Bode, Nyquist, Root Locus diagrams for a given control system and identify the parameters and carry out the stability analysis and to determine the frequency response of a control system and use it to evaluate or adjust the relative stability.
CO5	Design a P, PD, PI, or PID controller based on the transient and steady state response criteria and Model and analyse the control systems using state space analysis.